

MICROMOLD · DESIGN DECISION RECORD

MicroMold TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	MMD-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/micromold		
Source	docs/decisions/0001-trl2-review-decisions.md @ ff1ae73		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** decided. Items D1 to D10: Decided by Amish, 2026-09-25: go with recommendation (see MMD-DDR-002). Items O1 and O2 remain "Proposed, awaiting Amish".

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish", each with options and a recommendation, and MMD-PRC-001 v0.2 listed the same key design choices. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under that instruction and stays open for his review. Items without a recommendation stay open. Later on 2026-09-25 Amish wrote "i accept all your recommendations, go with them across all repos", so items D1 to D10 are now decided (MMD-DDR-002).

MicroMold uses none of the batch's shared components (FieldNode, CellGuard, MotionCore, ThermaCart, Twinkit, CalRig), so no cross-repo interface applies.

■ Options considered

The options for each item are those in docs/REVIEW.md (session 2026-09-25, /populate) and in MMD-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items decided.

#	Item	Decision
D1	Drive	Option (b): the rack, pinion and head of a 1 t arbor press on a taller steel column, with a screw press kept as a variant for small, thick parts. Decided by Amish, 2026-09-25: go with recommendation.
D2	Layout	Vertical barrel with the mold below. Decided by Amish, 2026-09-25: go with recommendation.
D3	Injection	A plunger rather than a reciprocating screw for the first build. Decided by Amish, 2026-09-25: go with recommendation.
D4	Heating	Two PID zones, barrel and nozzle. Decided by Amish, 2026-09-25: go with recommendation.
D5	Mold clamping	Bolted two-plate molds on a screw lift table, with a toggle clamp frame as a later upgrade. Decided by Amish, 2026-09-25: go with recommendation.
D6	Pressure indication	A 10 kN load cell under the ram rather than a spring-scale reading on the handle. Decided by Amish, 2026-09-25: go with recommendation.
D7	First mold	A 64 x 50 x 6 mm test plaque; product molds after co-design. Decided by Amish, 2026-09-25: go with recommendation.
D8	Budget	Option (a): keep \$400 and count molds as tooling outside the machine budget, reporting the mold cost separately. This redefines what the budget covers, so R14 in MMD-REQ-001 now reads "parts for the press, excluding molds, \$400 or less", and R8 covers the mold. <code>budget_usd</code> was later set to \$500 by MMD-DDR-002. Decided by Amish, 2026-09-25: go with recommendation.
D9	Fume control	Both: a small side hood with a duct fan in the BOM, and a written condition to run only under extraction or outdoors. Decided by Amish, 2026-09-25: go with recommendation.
D10	First co-design partner type	A group that already shreds HDPE or PP (for example a Precious Plastic workspace, a waste picker cooperative or a technical college that does). Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Items that remain open.

#	Item	Status
01	The specific first co-design partner and its city or region. D10 sets only the type of group; no partner or place was recommended.	Proposed, awaiting Amish
02	First product molds after the test plaque. These depend on what sells locally, which must come from co-design; no recommendation was made.	Proposed, awaiting Amish

■ Consequences

- `project.yaml`: only the TRL fields changed here. No pitch or problem rewording was recommended, and the rack-and-pinion drive is a lever-driven press, so the pitch ("a lever or screw press") still holds. `budget_usd` was \$400 at this record and is \$500 after MMD-DDR-002.
- MMD-PRB-001, MMD-PRC-001 and MMD-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed"; Amish decided them on 2026-09-25 (MMD-DDR-002). R14 is redefined (D8), and R13 now names the hood as well as the written condition (D9).
- The TRL 3 calculations (MMD-CAL-001) led to design changes within these items, decided with them by Amish on 2026-09-25 (MMD-DDR-002): a ratchet handle on the pinion (D1), a 4 mm nozzle orifice and a 5 to 7 mm

tapered sprue (D7), a glass-epoxy thermal spacer under the load cell (D6), a 20 mm bracket plate on mica pads, a nozzle zone shield and a nozzle tip raised to 190 mm so that 120 mm mold stacks fit (D5). With the hood, they bring the press to \$495, over the \$400 budget even under the redefined scope. New items raised by the calculations (budget, mass target, heater rating, mold cooling and feedstock grade) were listed in docs/REVIEW.md ; Amish accepted their recommendations on 2026-09-25, and MMD-DDR-002 records them and what changed.